

Ghosts, Geometry, and Reality in Fourth-Order Gravity

A guided introduction to physical claims, mathematical carriers, and evidence

GPT-5.6.sol*

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Abstract

Fourth-order gravity contains more local solutions than general relativity. Some appear with an indefinite sign in familiar decompositions and are called ghosts. The sign is a serious warning, but it does not identify by itself a physical particle. Local solution space, gauge-reduced classical phase space, nonlinear continuation, and positive quantum state space are different objects.

This paper develops a way to keep those objects apart. A physical conclusion is treated as a judgement depending on physical postulates, a mathematical carrier, logical and existence assumptions, and explicit bridges to observation. Pais–Uhlenbeck systems, compact pure-Weyl laboratories, Schwarzschild perturbations, Mannheim rotation curves, and the Bateman–Turok Euclidean lattice serve as controlled examples. Each answers a different question; none is silently promoted into a complete theory.

The examples yield exact survival results and exact obstructions. Additional classical pure-Weyl directions can survive gauge reduction. A repeated Schwarzschild spin-two block can form a non-split extension with a defective resonance. Strict pure Weyl has a nonzero local Euclidean one-loop BV anomaly in the declared complex. A full-complex BRST–Hadamard two-point pseudo-state can satisfy causal and gauge conditions while remaining indefinite. A bounded galaxy comparison can favor different models under different scores, and an explicit positive Euclidean family can refute a proposed universal stability shortcut.

The conclusion is methodological rather than triumphalist: the word “ghost” should be replaced by a ladder of falsifiable obligations, and each obligation should state which physics, mathematics, logic, and evidence make it true.

Authorship and how to read this paper

Asger Alstrup Palm, a computer scientist and Honorary Professor at DTU Compute, orchestrates the programme and is the accountable human contact. AI systems perform substantial derivation, programming, literature work, adversarial review, and drafting. Claude Fable 5 contributed to this introductory paper. Palm’s affiliation identifies his academic role and does not imply institutional endorsement.

AI authorship is neither evidence for nor evidence against a scientific claim. The relevant questions are whether the assumptions are declared, whether the derivation is valid, whether the computation can reject damaged inputs, and whether another researcher can locate the exact boundary of the claim. Sources, certificates, verifiers, and papers are available in the [public repository](#).

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1 An equation is not yet a theory

An elegant field equation does not say by itself what exists, what can be prepared, what evolves causally, or what an observer will count. A usable theory requires at least five layers:

1. **Physical postulates:** symmetry, locality, causality, conservation, and composition rules.
2. **A mathematical carrier:** finite algebra, Hilbert or Krein space, operator algebra, smooth fields, distributions, or an internal geometric setting.
3. **Logic and existence assumptions:** classical or constructive logic, forms of infinity, completeness, compactness, and choice.
4. **Operational bridges:** states, observables, probability, preparation, clocks, detectors, and reconstruction.
5. **Empirical contact:** a numerical prediction with declared inputs, uncertainties, and a falsifiable comparison protocol.

The programme writes this dependence schematically as

$$L + S + M + \text{Enc}(P) \vdash O,$$

where L is the logic, S the set or existence theory, M the mathematical carrier, P the physical postulates, and O the obligation being proved. The notation is not a claim that physics reduces to formal logic. It is a bookkeeping rule: if the carrier or existence principles change, the conclusion must be re-established rather than inherited by analogy.

Central rule. A finite identity is not a continuum limit. A Euclidean measure is not a Lorentzian causal state. A cohomology class is not a particle. A fitted curve is not a complete physical explanation.

2 The ghost question is four questions

A nondegenerate fourth-order oscillator has an equation of the form

$$\left(\frac{d^2}{dt^2} + \omega_1^2\right) \left(\frac{d^2}{dt^2} + \omega_2^2\right) z = 0.$$

It contains two second-order oscillatory branches. In the standard Ostrogradsky description, one normally appears with the opposite Hamiltonian sign [1, 2, 3]. Bender and Mannheim showed that the unequal-frequency Pais–Uhlenbeck model also admits a non-Hermitian PT-symmetric description with a positive metric [4, 5]. At equal frequency the branches coalesce into a Jordan system.

None of these facts settles gravity. In a gauge field theory the word “ghost” is commonly asked to carry four distinct meanings:

1. a local solution of a differential equation;
2. a classical direction surviving gauge symmetry, constraints, charges, boundaries, and the symplectic radical;
3. a direction admitting consistent nonlinear continuation;
4. a state or particle in a positive quantum theory.

The implications between these levels are neither automatic nor symmetric. A negative propagator residue warns of a problem but is not yet a gauge-reduced state. A nonzero classical symplectic pairing proves that a direction is not radical, but does not provide a Born rule. A positive metric for one free block does not guarantee an interaction-compatible metric. A well-behaved causal two-point distribution need not be positive on physical cohomology.

3 Why use several model universes?

No single background makes every obligation tractable. The programme uses declared laboratories, each chosen to expose a particular interface.

Laboratory	Question isolated
Pais–Uhlenbeck oscillator and scalar field	Positive and Krein completions, vacuum representations, Jordan limits, and interaction obstructions.
Conformal cylinder and compact product backgrounds	Full free gauge complexes, residual cohomology, causal transport, action-derived pairings, and nonlinear balance.
Schwarzschild exterior and static spherical metrics	Horizon and radiation conditions, extension structure, resonances, Green poles, reconstruction, and static charge normalization.
Mannheim–Kazanas geometry and NGC 3198	The separation between a fourth-order radial solution, matter and frame assumptions, fitted parameters, and an empirical score.
Bateman–Turok periodic Euclidean lattice	Whether a proposed all-field stability estimate survives an explicit positive, nonseparable counterfamily.
Finite, constructive, Hilbert, Krein, algebraic, and internal settings	Which physical obligations survive a change of mathematical foundation or carrier.

The laboratories are not asserted to be one universe. They are controlled experiments on the reasoning. A compact positive block does not establish an asymptotically flat particle; a Euclidean lattice does not establish Lorentzian reconstruction; and a black-hole endpoint theorem does not establish cosmology.

4 From oscillator signs to interacting fields

The elementary fourth-order models distinguish several structures often compressed into “the metric”: a finite-dimensional similarity transform, a Krein real form, a vacuum covariance, a field representation, a Fock sector, and the equal-frequency Jordan limit. These structures obey different continuity and ultraviolet requirements.

Linearized Einstein–Weyl gravity adds Lorentz covariance, helicity, gauge constraints, and a conformal critical limit. Gauge reduction removes redundant variables but does not turn the gravitational problem into independent copies of a scalar oscillator. The real form must be compatible with the tensorial and gauge structure.

Interaction is a separate gate. In the scalar and gravitational channels tested, resonant branch conversion obstructs the canonical analytic continuation of the free positive metric. The conclusion is scoped: it does not exclude every indefinite, nonlocal, singular, or nonperturbative prescription. It proves that a free completion cannot be advertised as an interacting solution without another theorem.

This distinction is especially important in higher-derivative gravity. Stelle’s renormalizability result [6] concerns ultraviolet power counting and counterterms; it does not by itself settle the physical state space. Conversely, a difficulty with one state-space completion does not erase the classical field equations or their gauge cohomology.

5 Pure Weyl gravity across the obligation ladder

5.1 Residual cohomology is not a graviton count

On the conformal cylinder, a declared zero-charge reduction treats all fifteen residual conformal generators as constraints. The first surviving centered classes are

$$H_{\text{res}}^4 = \text{span}\{[W_+^2], [W_-^2]\}, \quad G_{\text{res}} = I_2.$$

They are deformation or vertex classes, not one-particle gravitons. A companion causal BV–BFV construction transports the relevant free pairings between support conditions [7]. Neither statement supplies a positive interacting particle space.

Compact axial and polar laboratories show that some additional fourth-order directions survive gauge reduction and have nonzero action-derived Lee–Wald pairings. Their sign pattern is not the shortcut “Einstein positive, additional negative.” Other directions fail global charge balance, secular-growth, resonance, or Hamiltonian health tests. The scientifically stable object is therefore a classification of promotion and obstruction, not a universal rescue or universal no-go theorem.

5.2 The classical complex is an input to quantization

The classical BV–BFV complex is imported into the quantum programme through an immutable, content-addressed snapshot. Independent checks cover the declared nilpotency, contraction, chain-map, cyclicity, residual-cohomology, and pairing obligations. This gate matters because rebuilding an unofficial classical complex inside the quantum calculation could make a later quantum identity true of the wrong theory.

Typed q_2/q_3 compatibility with advanced and retarded Green homotopies then connects the nonlinear brackets to the causal classical data. The construction does not supply arbitrary all-order response trees, convergence, or an effective global Green solver.

5.3 Anomaly and pseudo-state are different quantum questions

In the declared local Euclidean quotient, strict fixed-field-content pure Weyl gravity has a nonzero one-loop BV anomaly. A formal shifting compensator makes the tested cocycle exact only after enlarging the field and counterterm space. That is a restoration in a changed theory, not proof that strict pure Weyl is anomaly-free.

On the Lorentzian side, the full 386-row off-shell carrier admits a BRST–Hadamard two-point pseudo-state pair satisfying the declared bisolution, graded commutator, wavefront, Ward, reality, and stationarity conditions. The word *pseudo-state* is essential. The pair is indefinite and does not establish positivity on physical cohomology, a positive graviton Hilbert space, a Feynman propagator, renormalized time-ordered products, a Lorentzian quantum master equation, scattering, or unitarity.

The anomaly and pseudo-state results do not repair one another. They answer different obligations on different dependency rails.

6 Black holes: boundary selection and defective response

Maldacena-type conditions can isolate an Einstein sector of conformal gravity in Euclidean anti-de Sitter or late-time de Sitter settings [8]. The Lorentzian Schwarzschild question is different: do

future-horizon regularity and ordinary one-sided radiative traces force the additional pure-Weyl curvature carrier to vanish?

For the complete axial $\ell = 2$ Schwarzschild system, the answer is no. Its radial differential module has successive factors

$$M_{\text{RW}}^{(2)}, \quad M_{\text{RW}}^{(2)}, \quad M_{\text{RW}}^{(1)}.$$

The repeated spin-two factors form a non-split self-extension rather than a direct sum. Exact Wronskian arguments populate every incoming direction by a future-horizon-regular exterior solution for real positive frequency. The same filtration excludes exponentially growing separated axial modes.

This coexistence is instructive. Separated-mode stability says that one particular instability is absent. It does not make the inherited flux positive and does not bound the full time-domain resolvent.

At one validated simple Regge–Wheeler quasinormal frequency [9, 10], the complete connection has Smith valuations

$$(0, 0, 2).$$

The algebraic multiplicity is two, the geometric multiplicity is one, and the root chain has length two. The geometric root lies in the Einstein submodule; the generalized root has nonzero gauge-invariant traceless-Ricci carrier. This is related to the logarithmic-partner structure of critical gravity [11, 12], although the asymptotically flat endpoint tangent is polynomial rather than an independent literal $\log r$ mode.

The compactly cut-off exterior radial Green operator has

$$R(\omega) = \frac{R_{-2}}{(\omega - \omega_n)^2} + \frac{R_{-1}}{\omega - \omega_n} + R_{\text{hol}}(\omega), \quad R_{-2} \neq 0.$$

The same pole appears in a fixed-domain two-ended complex-scaled Fredholm realization and is the meromorphic continuation of a retarded, mode-reduced, compact-source/compact-observation transfer. An isolated local contour therefore contains $e^{i\omega_n t}(V_1 + itV_0)$.

Boundary of the resonance claim. The non-split extension, double radial Green pole, and retarded mode-reduced parent are established in their declared settings. A full metric/BV retarded propagator, global inverse-Laplace deformation, complete late-time expansion, and real astrophysical detector waveform are not.

Static Bach-flat black holes answer another question. Residual-basic normalization yields a consistent Hamiltonian, Wald entropy, and signed temperature on the certified static family, but this is not a radiative or physical-process thermodynamics theorem.

7 Two tests beyond black-hole perturbation theory

7.1 Mannheim gravity: solution, interpretation, and fit

The Mannheim–Kazanas geometry contains a linear radial potential because the static fourth-order vacuum equation admits it [13, 14]. That mathematical fact does not determine how the term is sourced or how massive matter moves. A physical explanation must also specify the matter sector, scale breaking, conformal frame, source moments, and global boundary data.

The distinction becomes visible in a bounded 39-point NGC 3198 protocol:

Model	RMS (km/s)	Reduced χ^2	Random-error gate
Newtonian baryons	23.896	128.722	fail
GR plus NFW halo	5.148	0.965	pass
Mannheim conformal gravity	4.694	3.202	fail

Mannheim has the smallest unweighted residual, whereas GR+NFW has the better uncertainty-weighted score. This is not a contradiction: the two scores ask different questions. Nor is it a population-level verdict. Distance, inclination, beam smearing, stellar populations, gas profiles, and other systematics lie outside the frozen protocol.

7.2 Bateman–Turok: a universal shortcut meets a counterfamily

Bateman and Turok propose a positive Euclidean lattice and a hidden ghost parity as a route around the usual higher-derivative instability story [16]. One tempting deterministic strategy is to bound the complete action gradient below at the free bilaplacian scale for every positive field.

An explicit periodic-torus family disproves that universal bound. On $L_n = n^{24}$, an exactly mean-compensated critical source is inverted by the periodic graph Green operator, lifted above its minimum, and coupled weakly in four directions. The fields remain positive, nonseparable, and of polynomial contrast. Their action stays bounded above and away from zero, while

$$\frac{Q_n}{\omega_{L_n}^2} = O(n^{-2}) \longrightarrow 0.$$

This refutes one deterministic all-field scaled-PL architecture, not the full Bateman–Turok programme. Gibbs typicality, the interacting Witten quotient, the H^{-1} moment, the continuum measure, reflection positivity, and Lorentzian reconstruction remain logically separate tests.

8 Reverse foundations: changing the mathematical world

The previous examples change physical models while mostly retaining familiar mathematics. Reverse foundations asks the complementary question: what happens when the physical obligation is held fixed but the mathematical or logical background changes? This extends the idea of reverse physics—work backward from what a law must accomplish to the assumptions that make it possible [15].

The interactive atlas crosses three axes:

- six foundational regimes, including classical, weakened-choice, constructive, internal/topos, and finite settings;
- six carrier families, including finite algebra, Hilbert space, Krein space, operator algebra, smooth/distributional fields, and localic geometry;
- sixteen physical obligations, from state existence and dynamics through causality, gauge reduction, interaction, renormalization, reconstruction, and empirical adequacy.

The Cartesian product contains $6 \times 6 \times 16 = 576$ coordinates. They are not 576 candidate universes. They are 576 scoped questions. A populated cell records evidence about one obligation in one setting; it does not show that neighboring cells compose.

Theory assemblies group evidence by research tradition, while theory passports follow a single route through foundations, state space, dynamics, observable, prediction, and empirical benchmark. This exposes an important difference between *coverage* and *completion*. A mature tradition may possess tools for every job without those tools belonging to one model. An unconventional programme may solve a difficult late-stage identity while lacking an earlier state or observable bridge.

Finite and constructive examples can avoid particular basis-selection or completion assumptions. Lean/Physlib proof passports can replay finite identities with explicit types and signs. Such results show that an assumption is avoidable in that representation. They do not prove that the universe is finite, that all physics is constructive, or that formal proof supplies a missing physical premise.

9 Evidence is part of the theory interface

To prevent scope changes from becoming invisible, quantum results carry one or more exact dependency tags:

LOCAL-ALGEBRAIC EUCLIDEAN-SPECTRAL REDUCED-MODE LORENTZIAN-CAUSAL.

A result does not move between these rails without an explicit bridge. Likewise, anomaly classification, coefficient computation, QME restoration, residual transfer, and Lorentzian certification are distinct lifecycle states.

The preferred audit chain is

claim $\longrightarrow$ producer $\longrightarrow$ certificate $\longrightarrow$ independent verifier $\longrightarrow$ adversarial test $\longrightarrow$ paper.

Re-running the producer is reproduction, not independent verification. Exact algebra and numerical validation remain different evidence types. A timeout, missing input, or skipped exhaustive calculation is not a pass.

This architecture is especially important for AI-orchestrated research. AI systems can derive equations, write software, search literature, and produce fluent exposition. They can also share hidden assumptions, invent references, and produce mutually reinforcing errors. The answer is not to treat AI as an oracle, but to make every substantial claim easier to attack. A reproducible refutation is scientifically useful because it removes an invalid route and sharpens the surviving alternatives.

10 What the examples establish

Six conclusions cut across the different laboratories:

1. **A sign is not a state.** Pole residues, classical fluxes, Krein signatures, and quantum probabilities are different objects.
2. **Reduction can preserve or remove extra directions.** Gauge, charges, boundaries, and the symplectic radical must be tested explicitly.
3. **Interactions are promotion gates.** A free positive completion can fail when branches mix, while a formal nonlinear solution can fail global balance or boundedness.

4. **Causality and positivity are independent obligations.** A BRST–Hadamard pseudo-state can have the required singularity and gauge identities without defining a positive physical state.
5. **A counterexample should be scoped as carefully as a theorem.** The BT family refutes one all-field bound; the NGC 3198 gate tests one frozen data protocol; neither decides an entire research tradition.
6. **Changing mathematics changes proof obligations.** Replacing a Hilbert carrier, weakening Choice, or restricting to a finite model can remove an assumption, but operational meaning must be rebuilt rather than presumed.

These conclusions explain why collecting individually attractive ingredients is not enough. A physical theory requires typed interfaces between its foundations, states, dynamics, observables, predictions, and empirical tests. The interfaces are often where the decisive theorem or obstruction lives.

11 Routes into the technical evidence

The [Reverse Physics Atlas](#) provides the interactive matrix, dimension guide, programme assemblies, theory journeys, Weyl BV routes, literature records, certificates, and formal-proof passports. The companion papers can be entered thematically:

- [the four-level ghost classification](#) develops the separation between local, classical, nonlinear, and quantum claims;
- [the Mannheim analysis](#) separates the radial solution from matter, frame, boundary, and empirical premises;
- [the gauge-algebra analysis](#) asks whether algebraic structure is input, derived, or representation-dependent;
- [the reverse-foundations synthesis](#) develops the atlas and literature framework;
- [the Bateman–Turok analysis](#) proves the compensated Green-tail counterfamily.

These routes are thematic references, not a sequence of release notes. Each technical claim should be evaluated through its own assumptions, certificate, verifier, and stated non-claims.

12 Conclusion

Fourth-order gravity cannot be assessed by counting poles or attaching the word “ghost” to every additional solution. The physically relevant question is which object survives each promotion gate: local equation, classical reduction, nonlinear continuation, causal construction, positive quantum state, operational prediction, and empirical test.

The examples do not establish pure Weyl gravity, Mannheim conformal gravity, or Bateman–Turok physics as a successful description of nature. Nor do they support a single generic dismissal. They establish a mixed landscape of survival theorems, obstructions, counterexamples, and unproved interfaces.

Durable conclusion. Physical postulates, mathematical carriers, logical strength, causal and quantum claim types, interface proofs, and empirical gates must be separated. Only explicit bridges may join them.

That principle turns a vague dispute about ghosts into a collection of answerable questions. It also gives researchers a practical standard for reviewing theories built with unfamiliar mathematics or substantial AI assistance: inspect the obligation, inspect its dependencies, and inspect the bridge to the next claim.

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